

Advanced Computer Graphics

Animating a Human

Virtual Human

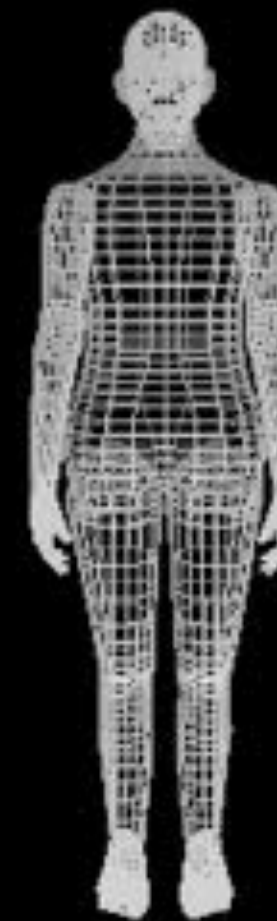
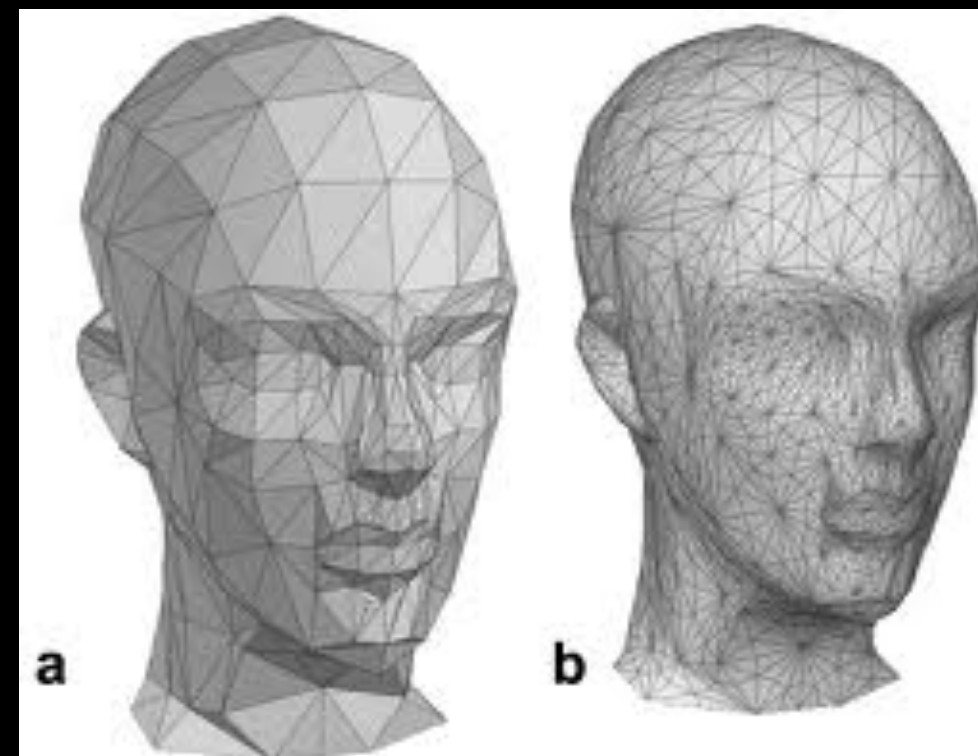
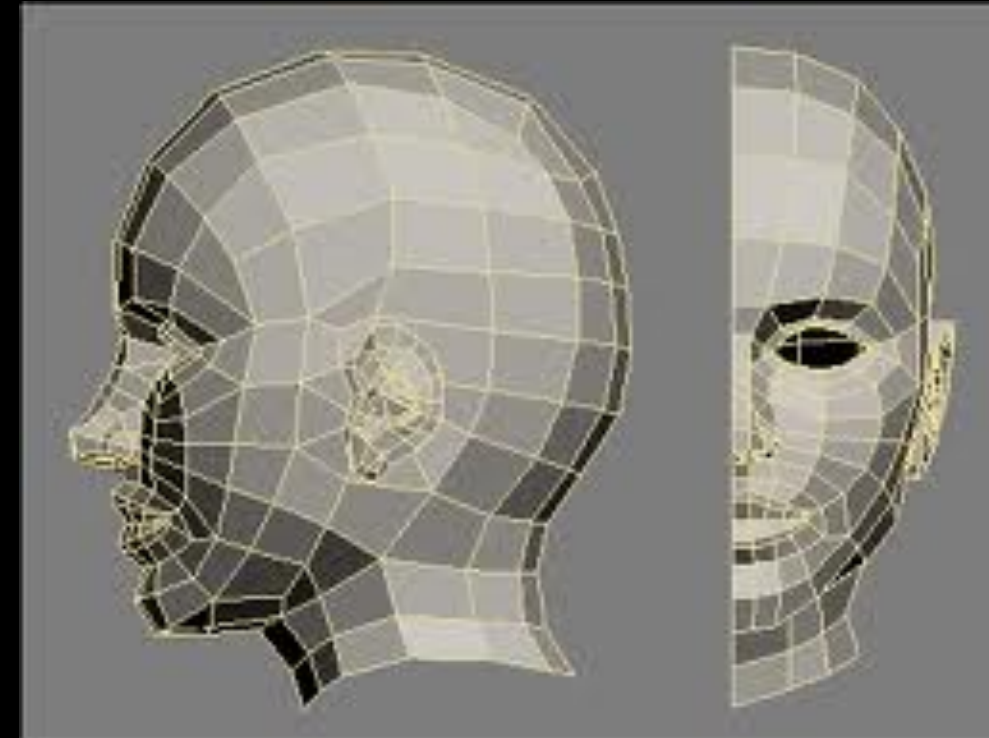
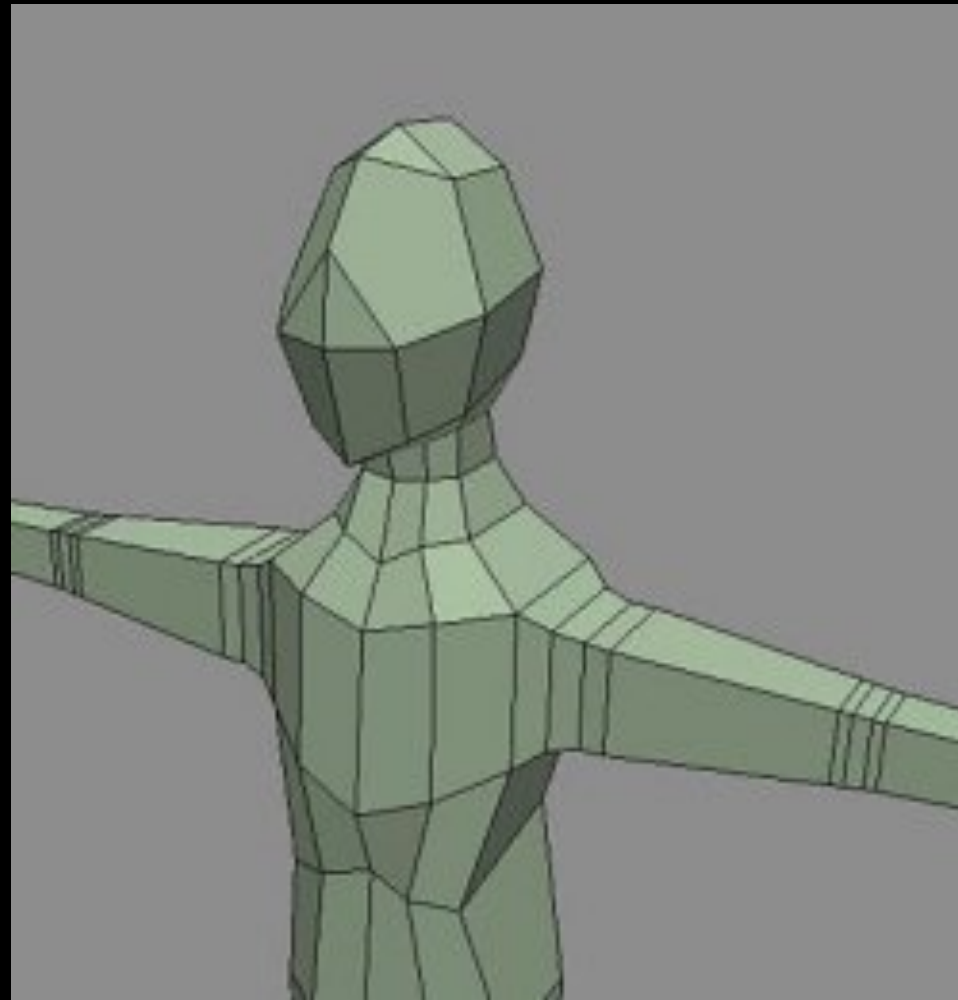
- Virtual Human
 - Body Modeling
 - Behavior Modeling

Body Modeling

- Constraints
 - Real-time Rendering
 - Animatable

Body Modeling

- Shape in Polygons

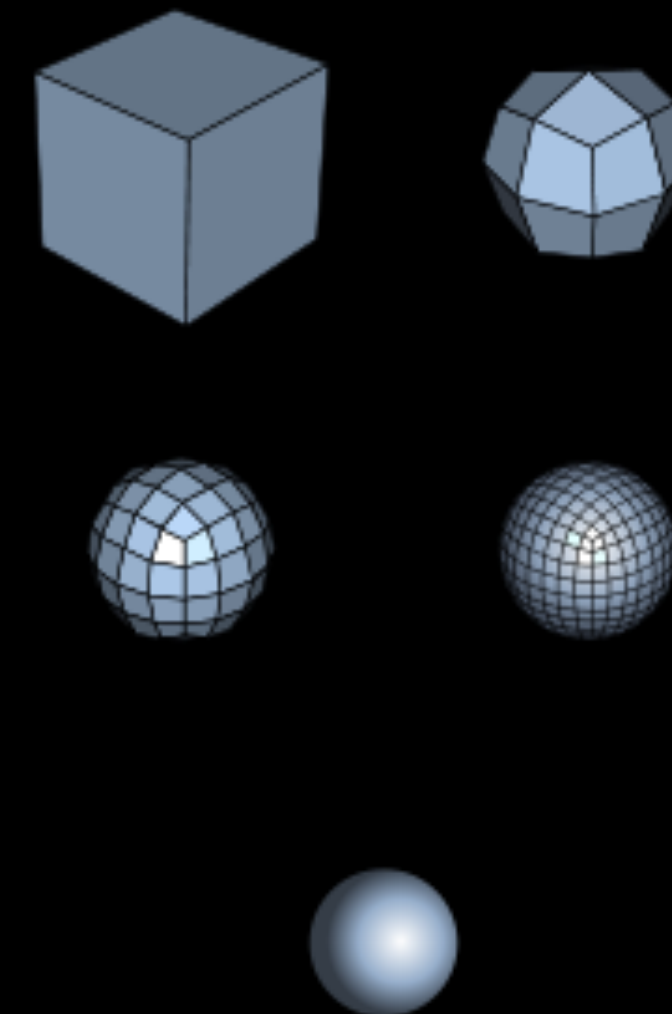
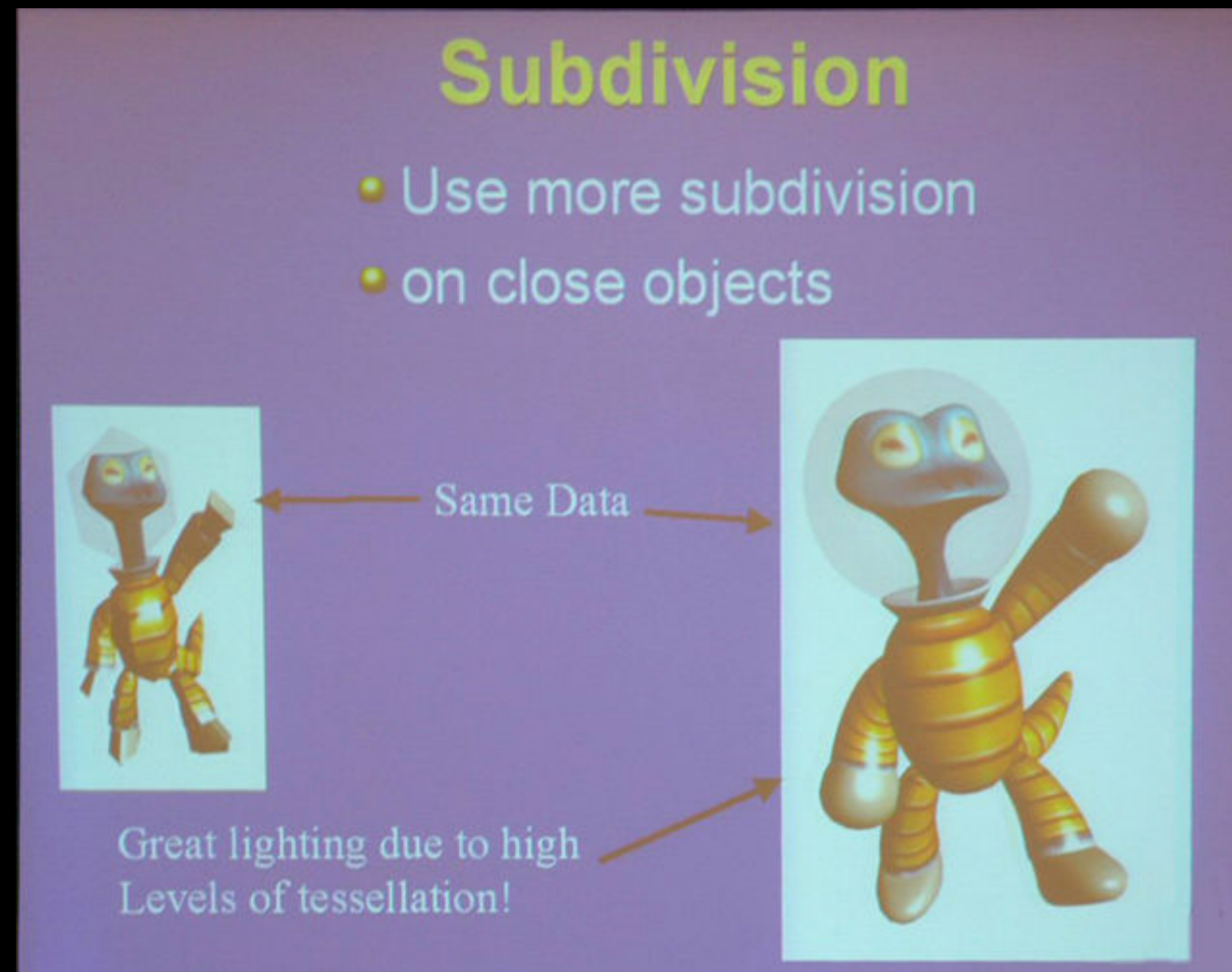


Body Modeling

- Controlling Number of Polygons
 - Why?
 - Real-time Rendering vs. Accuracy

Body Modeling

- Using Subdivision Surfaces



Body Modeling

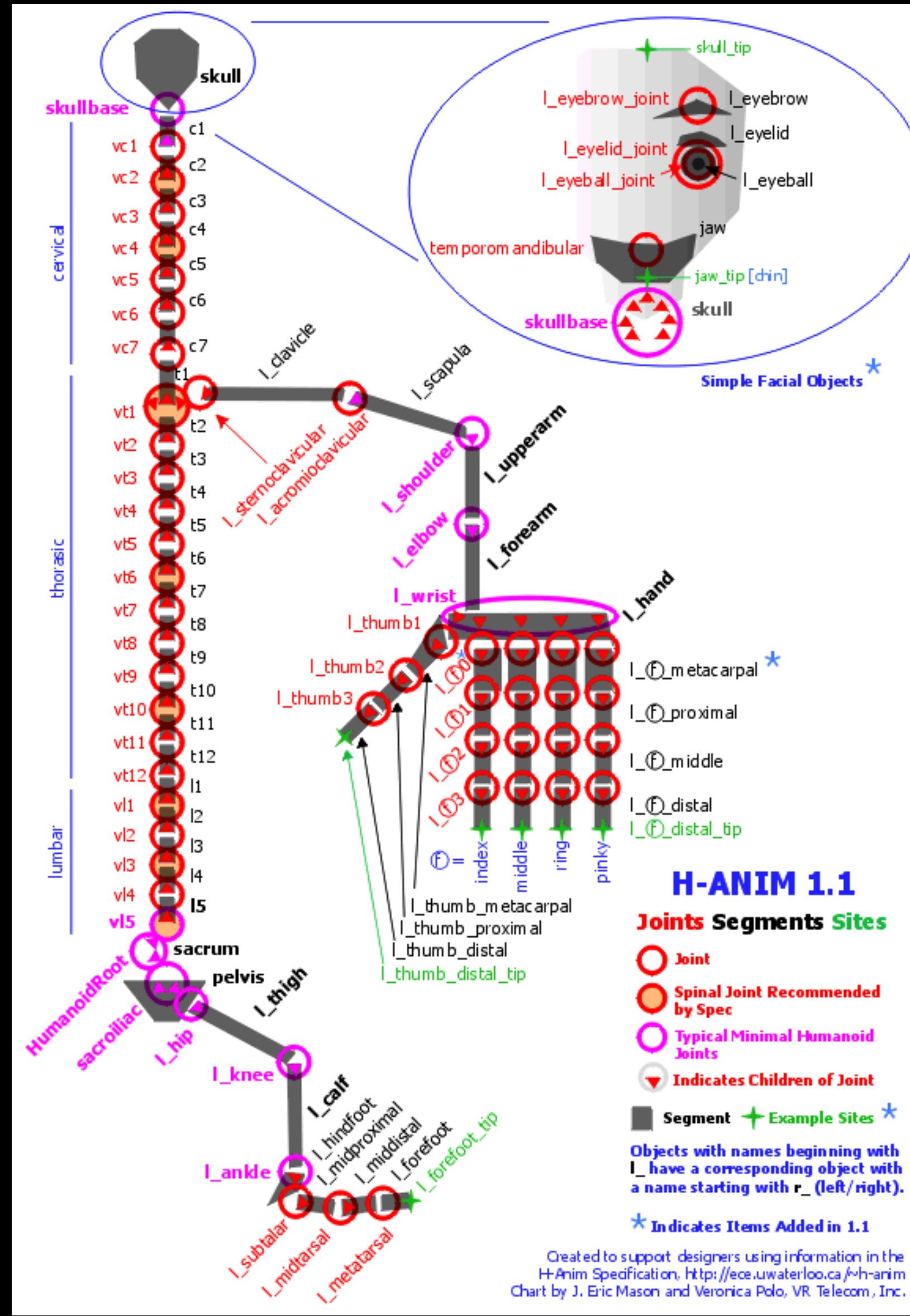
- How to get the original shape? aka the most complex shape



Standard Structure

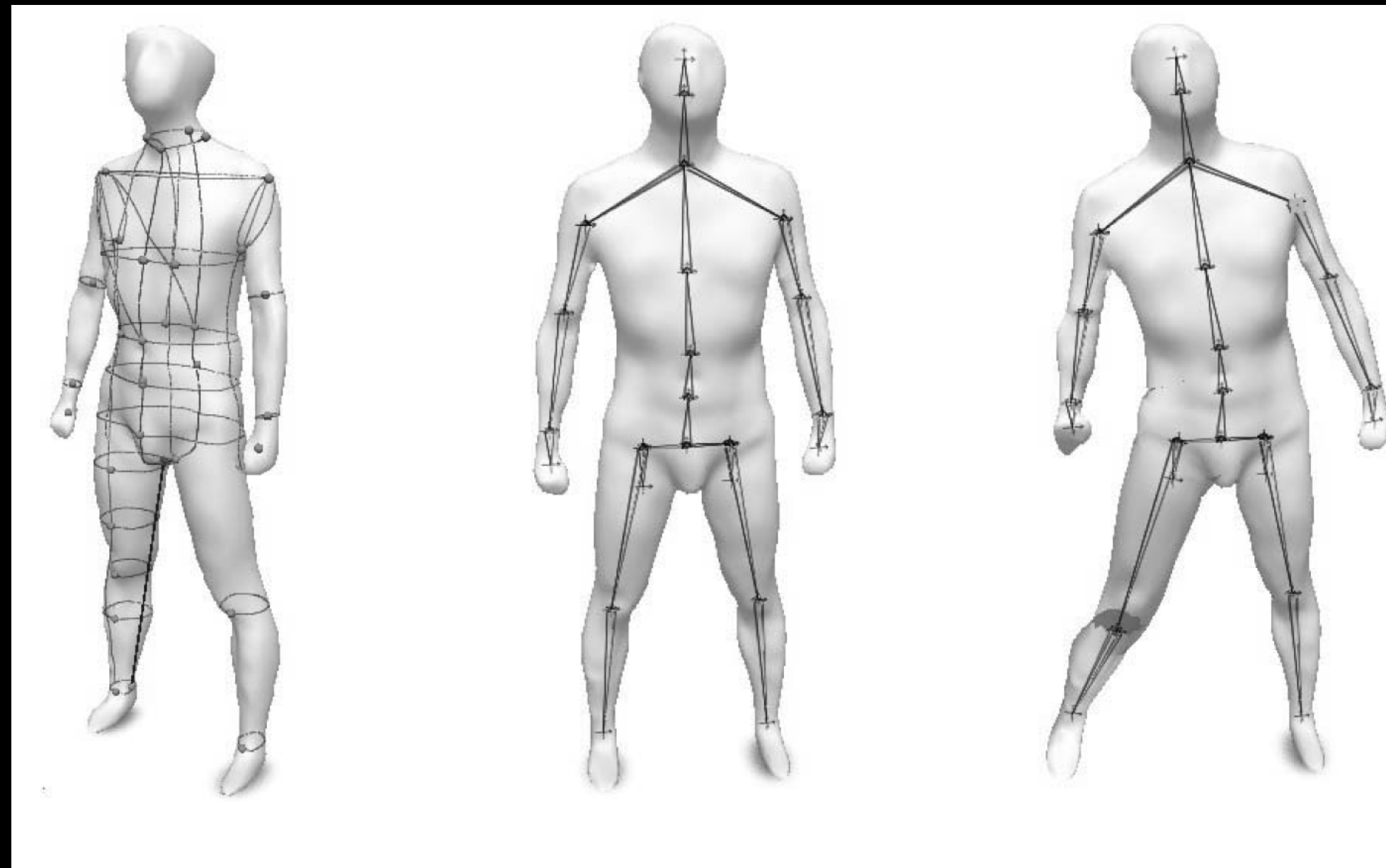
H-Anim Structure

- Why standard?
- Joint by name



Body Modeling

- Create Animatable Structure
 - Joints and Links
 - + Skins



Rigging Examples

Pinocchio

Automatic Rigging and Animation of 3D Characters

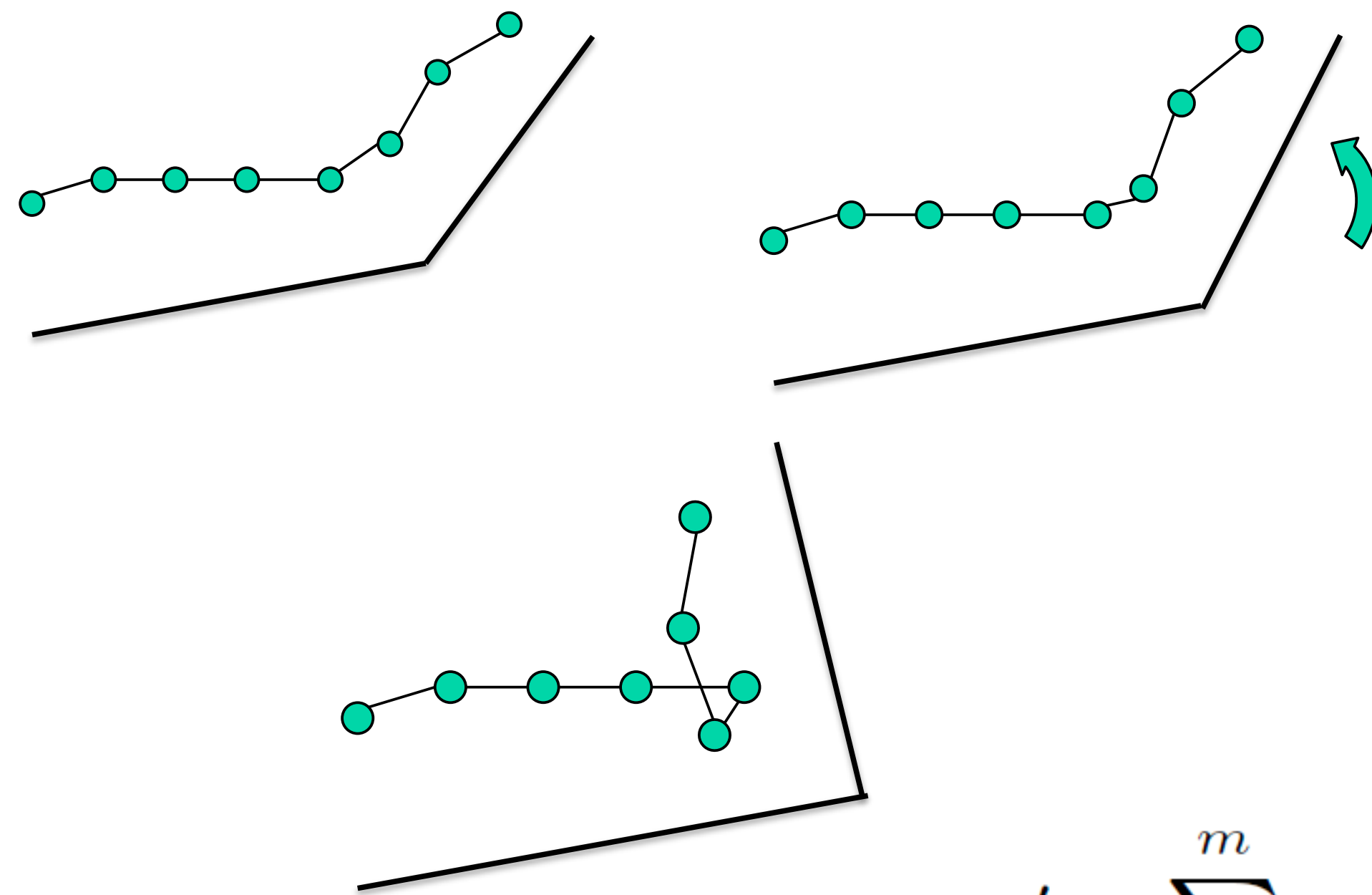
Ilya Baran and Jovan Popović

SIGGRAPH 2007

papers_0030

Body Modeling

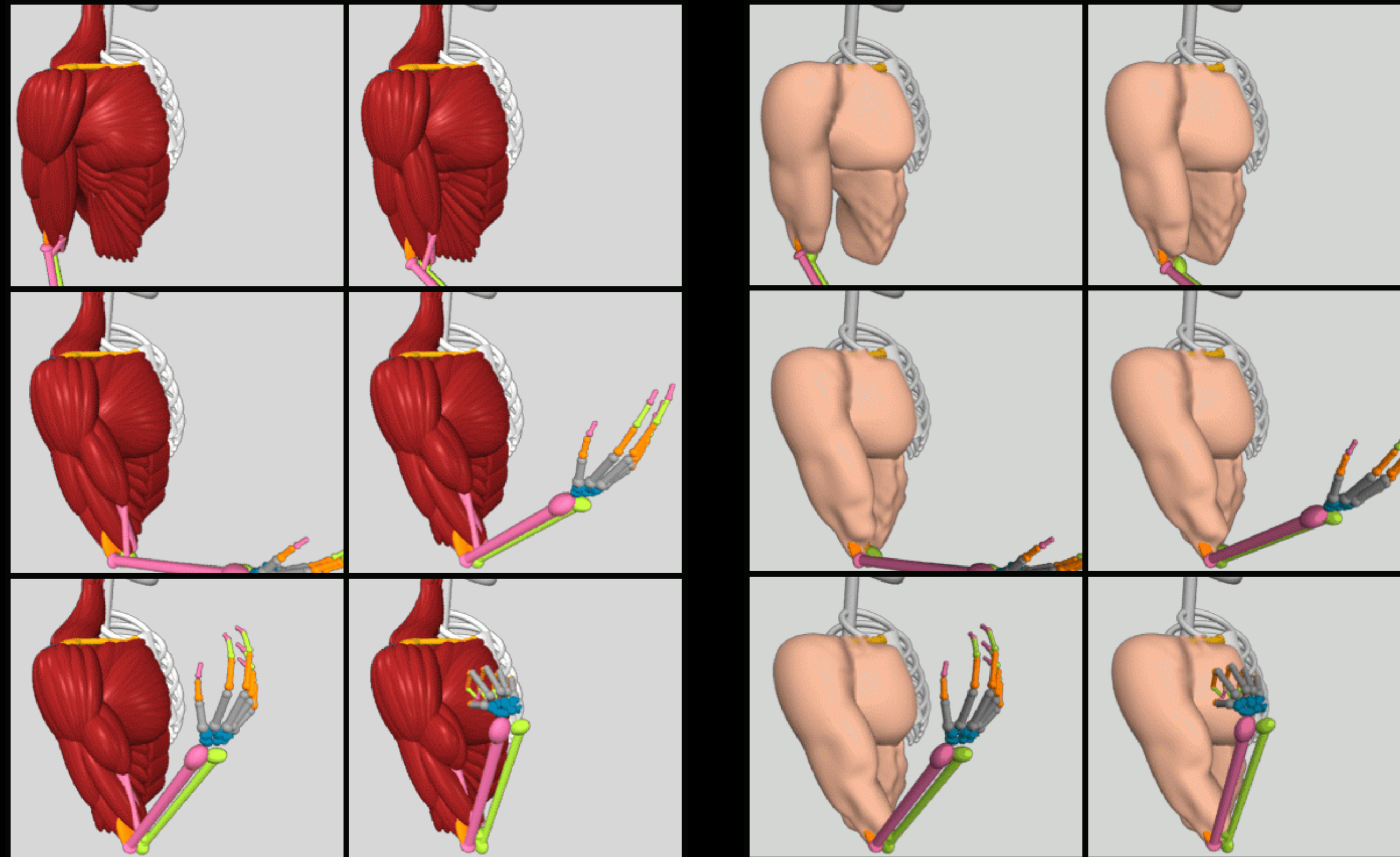
- Skinning



$$\mathbf{v}'_i = \sum_{j=1}^m w_{i,j} \mathbf{T}_j \mathbf{v}_i$$

Body Modeling

- More accurate model?



Summary

- Body modeling
 - Shape = Skeleton + Skin
 - Animation = Skeleton based animation
- Animation
 - Motion captured

Okay

- Now we have animatable body in general
- How to control it?
- Behavior issues - the structure and shape should be modeled to be adopted for target behavior

Controlling Animatable Body

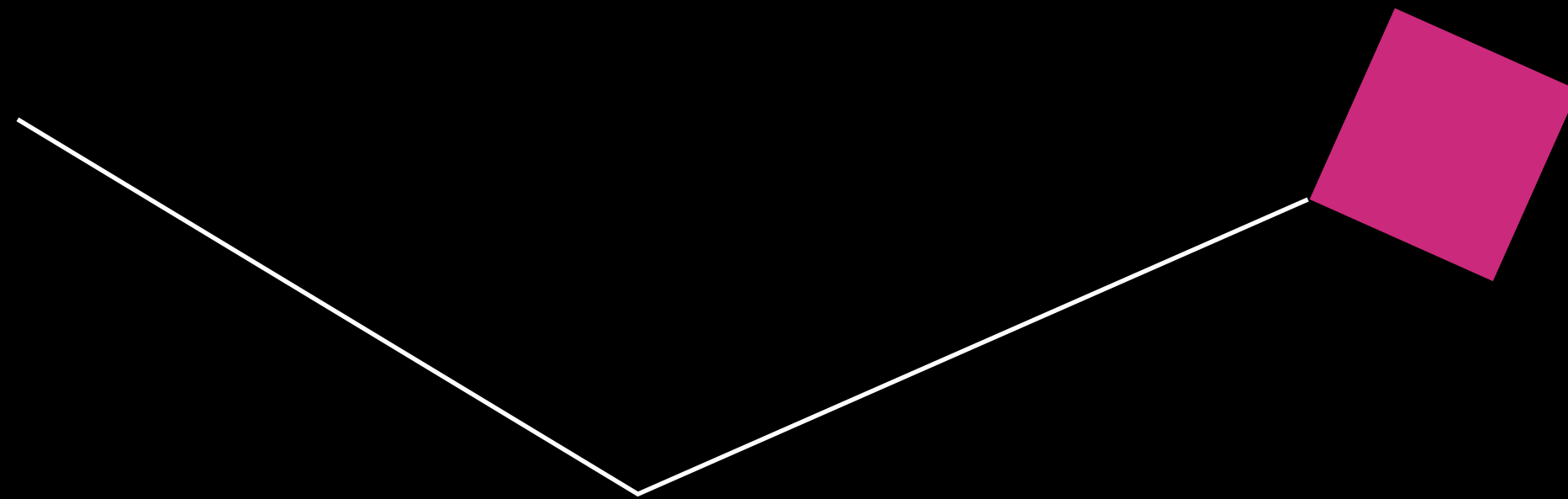
- Case 1) Upper body movement
 - Reaching

Modeling Reaching Motion

- step 1: Arm should be modeled to be able to animate reaching motion

1. Modeling the Arm

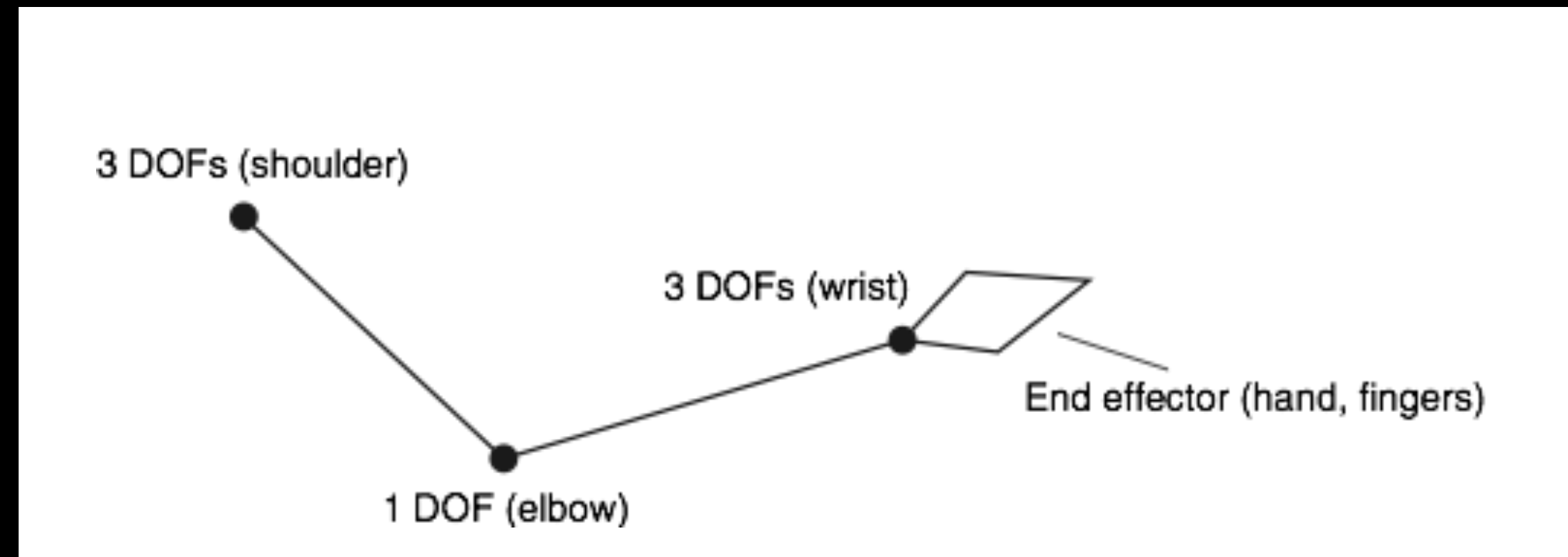
- Joint and Links



- DOF for each joint?

1. Modeling the Arm

- Case 1: 3-1-3 DOF
- Case 2: 3-2-2 DOF



2. Control Mechanism

- Okay, then how to specify the reaching animation?
- Specifying goal position!
- of what?
 - Finger tips?
 - Wrist position

2. Control Mechanism

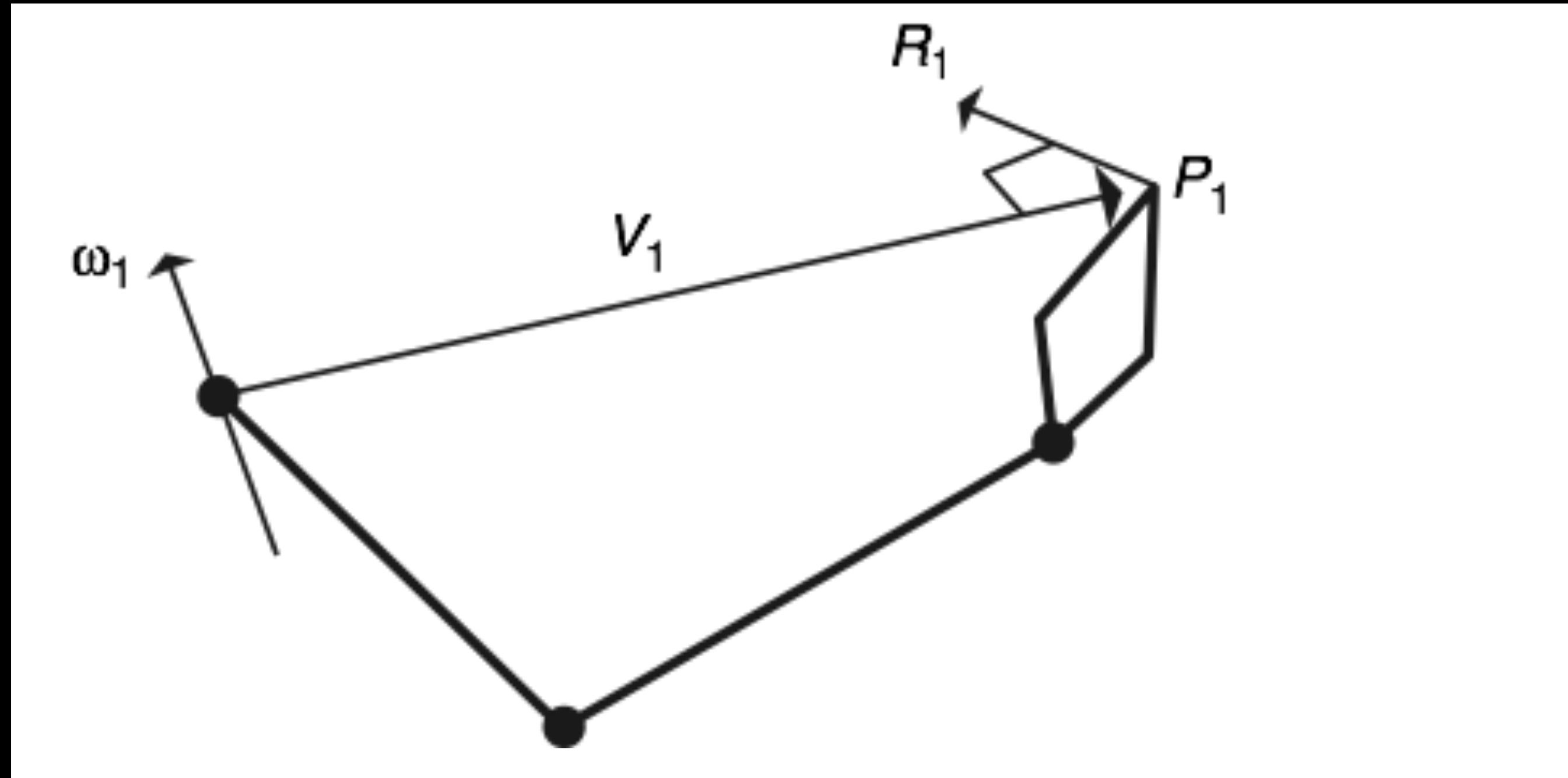
- For reaching animation
 - Fix wrist position as a goal position
 - IK is under-constrained problem even with inverse Jacobian method
 - How to solve it?

2. Control Mechanism

- Inverse Jacobian method considers all joints at equal importance
- How about human motion?
 - Base joint - gross motion
 - End joint - fine motion
- Procedural approach!

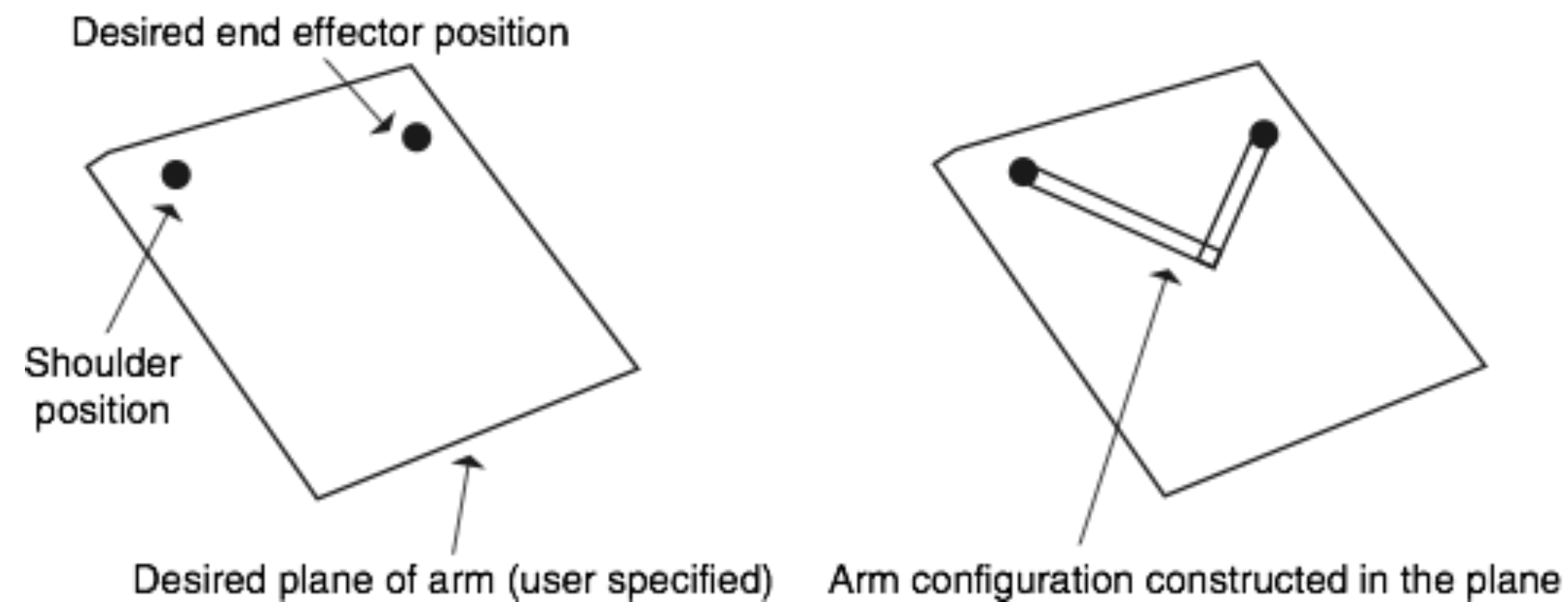
2. Control Mechanism

- Effect of each joint to the end effector



2. Control Mechanism

- Still too much DOFs!
- Reduce DOF by specifying plane of arm movement

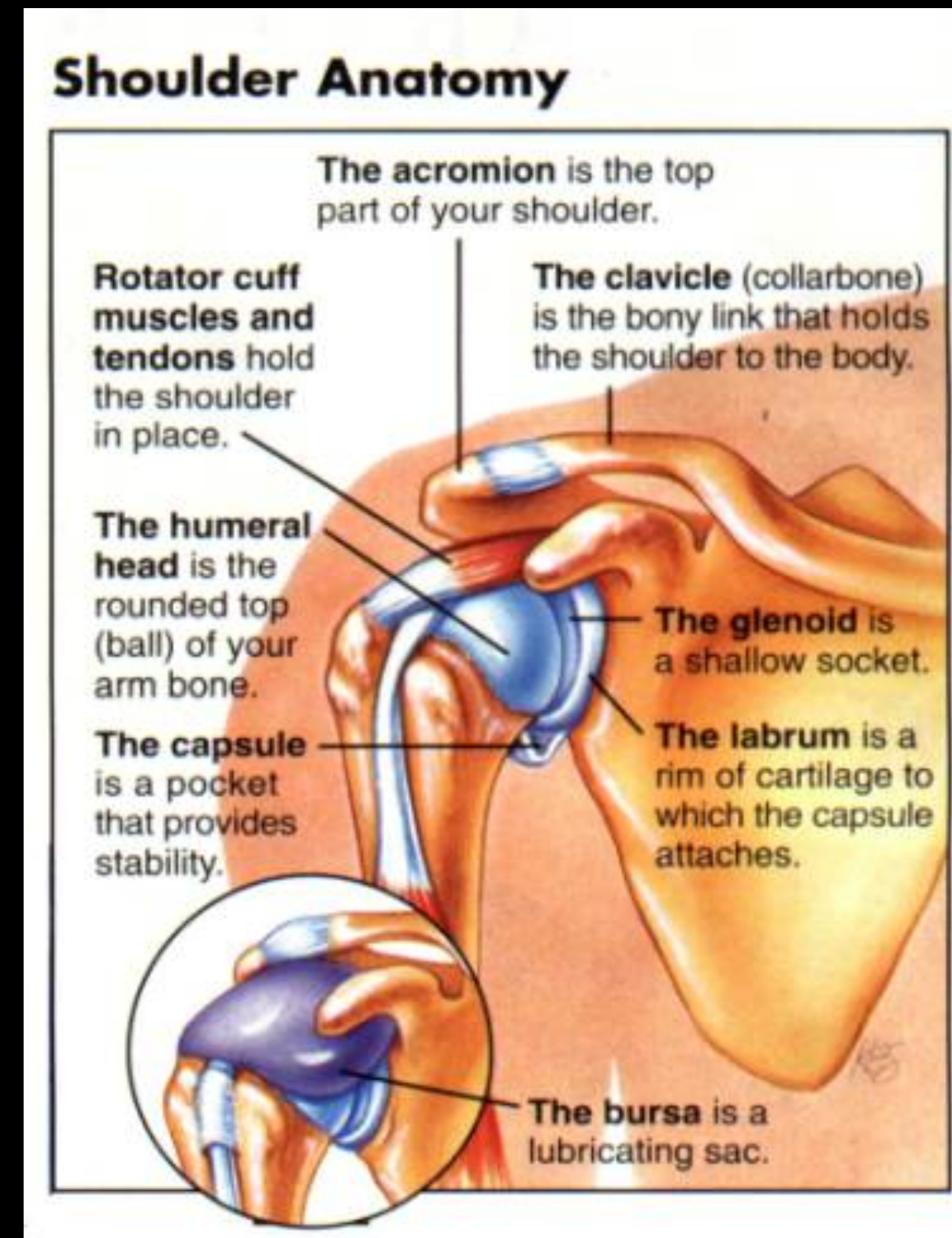


2. Control Mechanism

- With fixed shoulder and wrist, reaching animation is reduced into 1 DOF problem.
- Wrist DOFs are controlled in fine level to actually reach to the position: 2 DOFs
- How about shoulders?
 - 3DOFs!

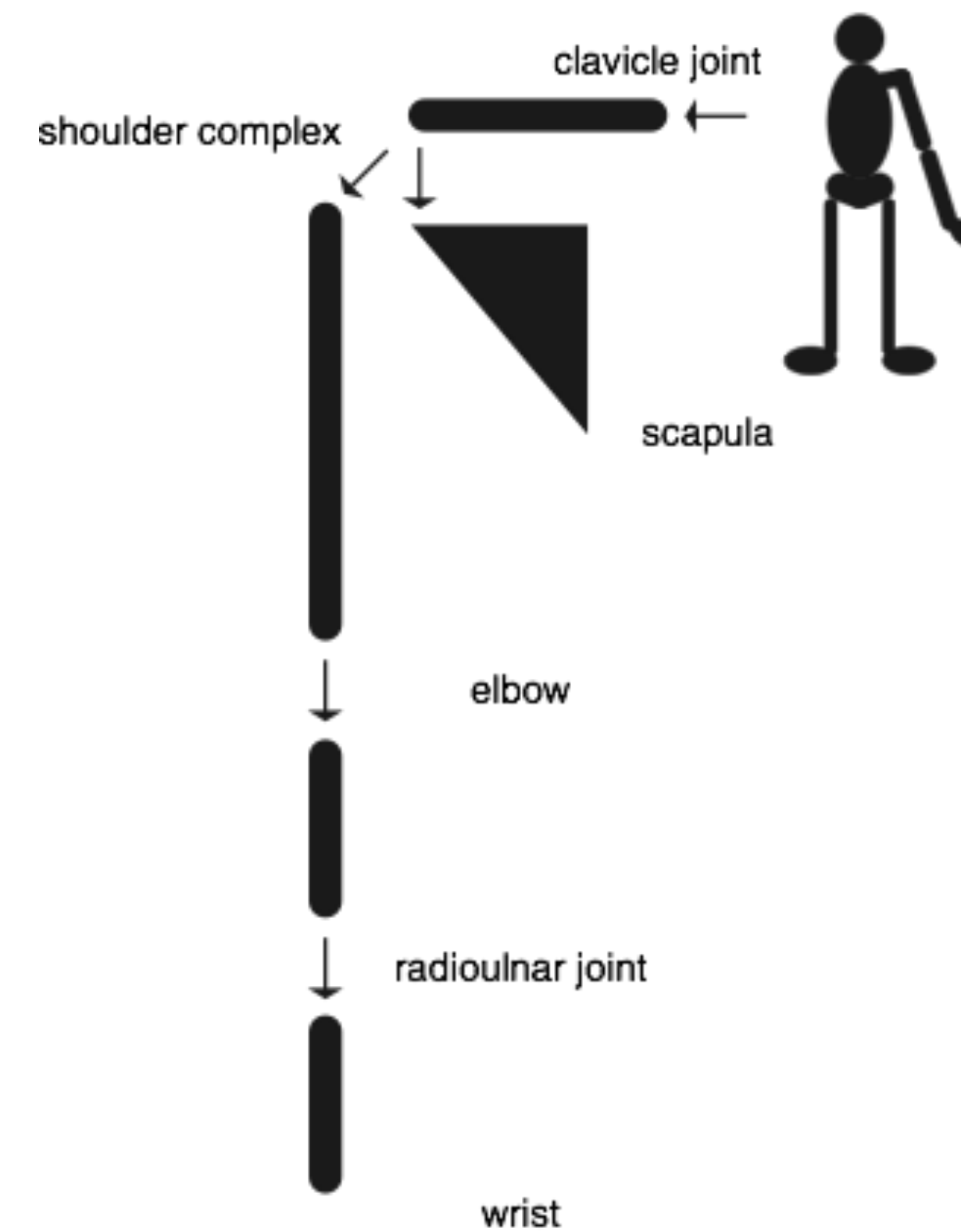
3. Modeling Shoulder Joint

- Why shoulder?
- 3DOF!
- Complex!
 - One of the most complex human joint



3. Modeling the Shoulder

- Simple case: 1 3DOF joint
- Realistic case: With 3 joints



3. Modeling Shoulder Joint

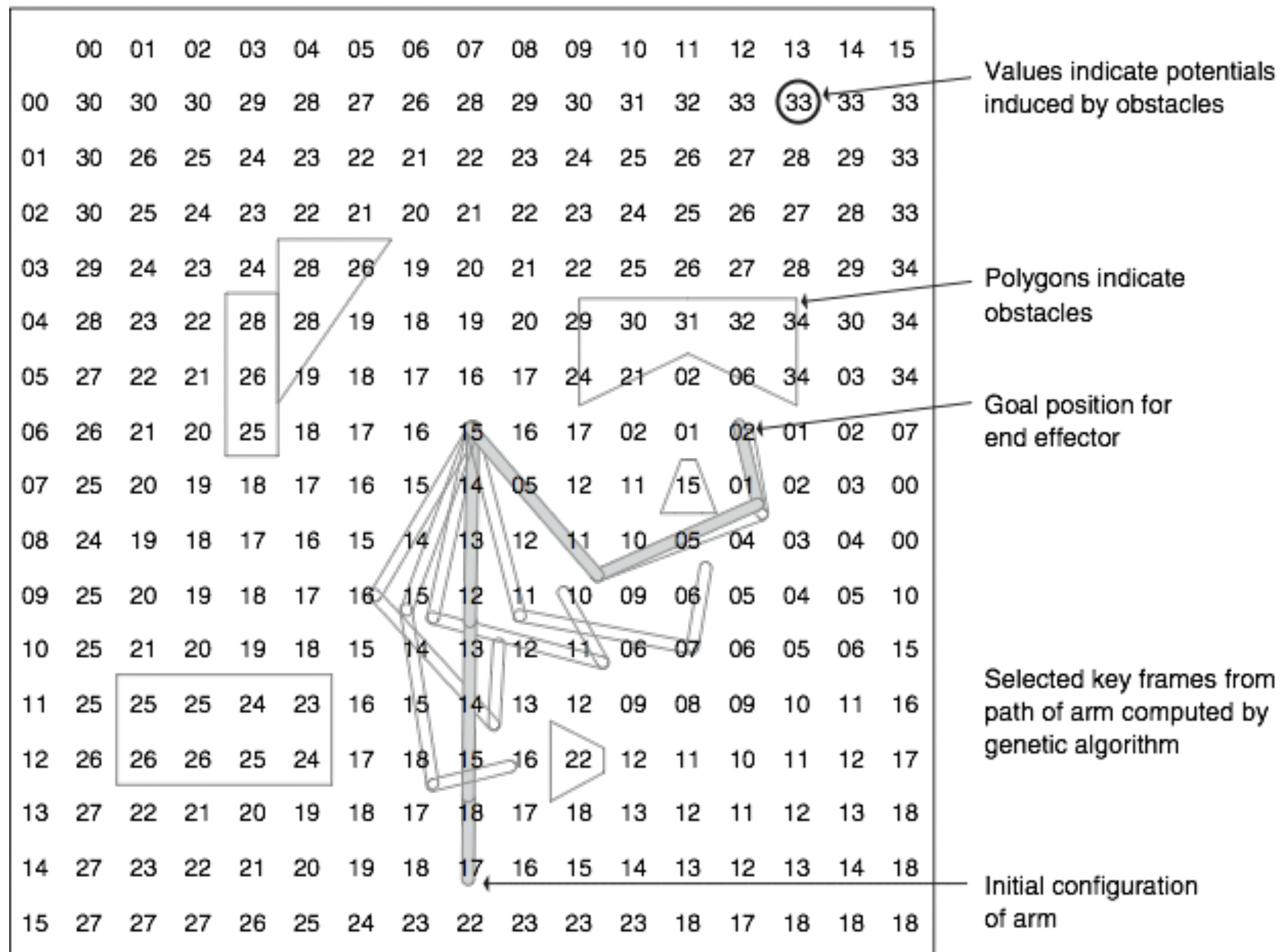
- Animating Shoulder
 - Usually animated to reach motion plane
- Overall step
 - Define goal position
 - Define intermediate motion plane
 - Determine shoulder movement
 - Determine elbow movement
 - Determine wrist movement

4. Reaching around an obstacle

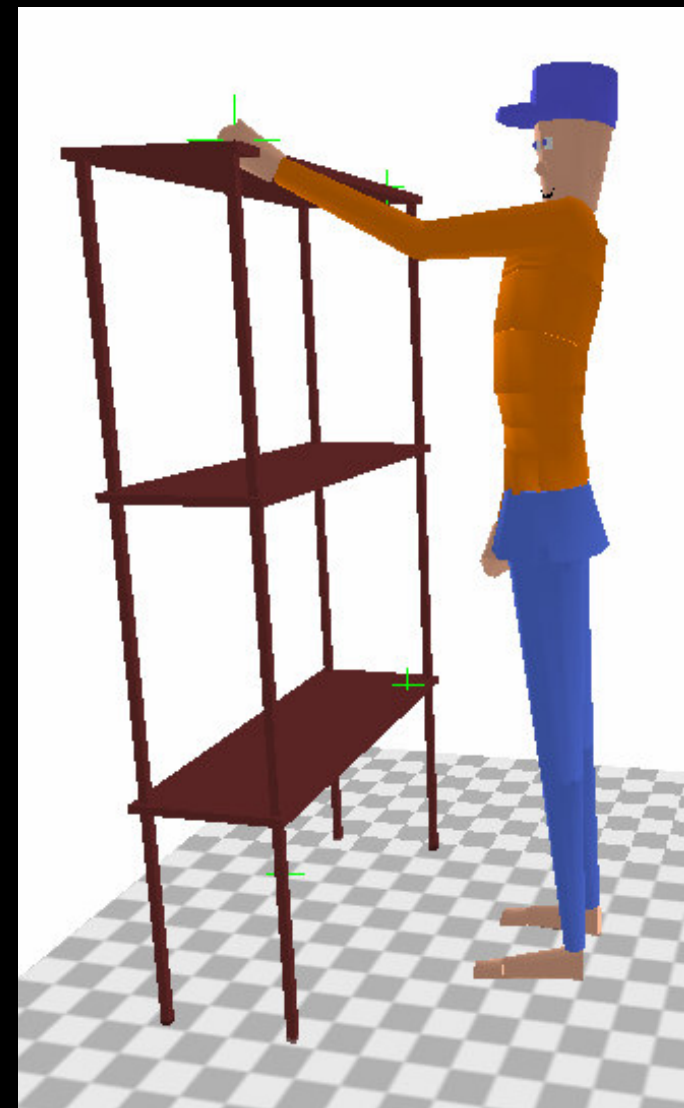
- How to reach to a position over obstacles?
 - One obstacle - step-by-step procedure
 - Many obstacles?
- Modeling environments

4. Reaching around an obstacle

- Case study: Potential fields
 - Obstacles: impacts positive repulsion energy
 - Goal: impact negative repulsion energy
 - Movement: following largest gradients



4. Reaching around an obstacle

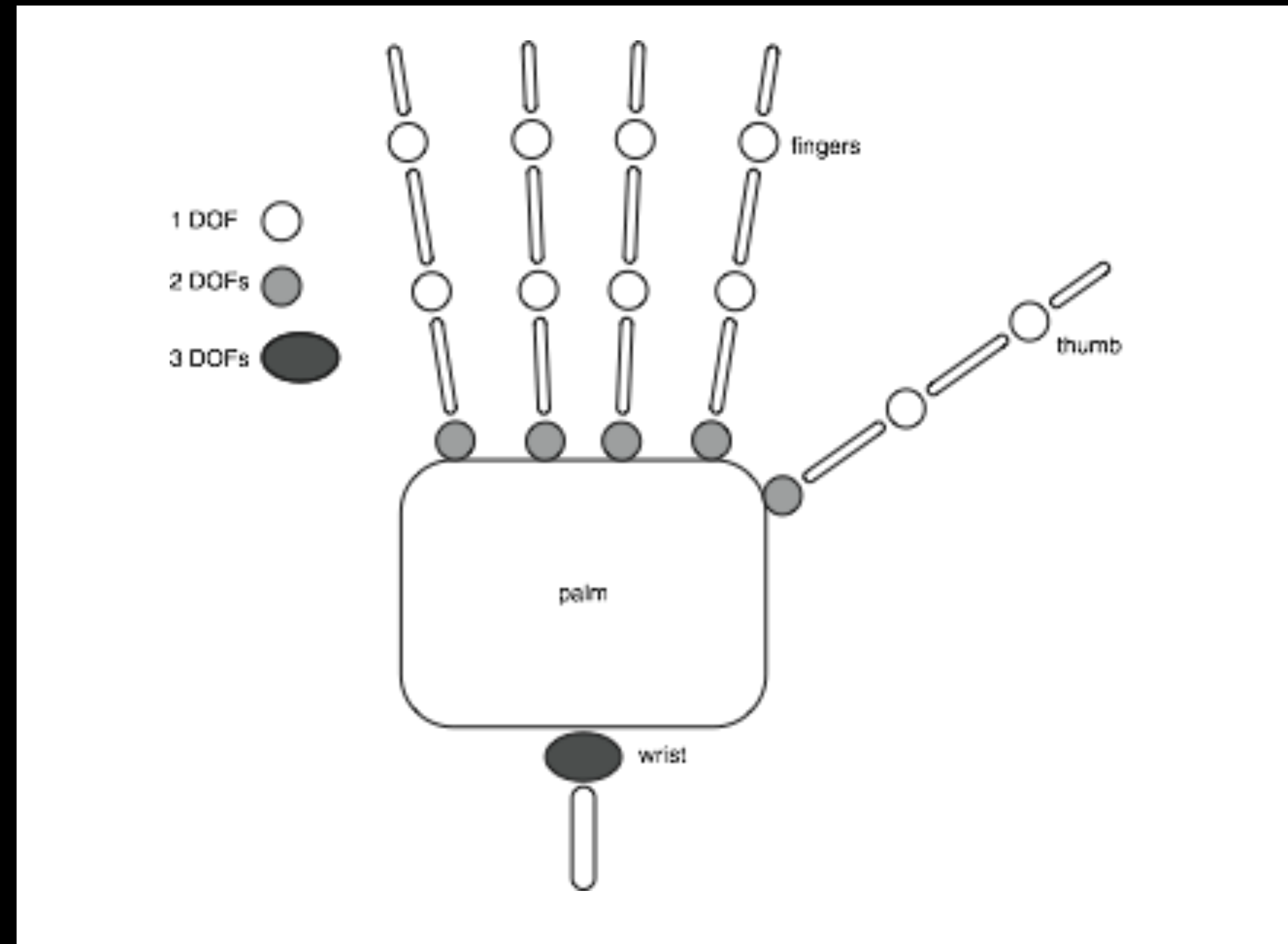


5. Modeling an Hand

- Final step!
 - After reaching we need to grasp the object
 - How to animate?
 - Is it same procedure with reaching?
 - No!

5. Modeling an Hand

- Complex

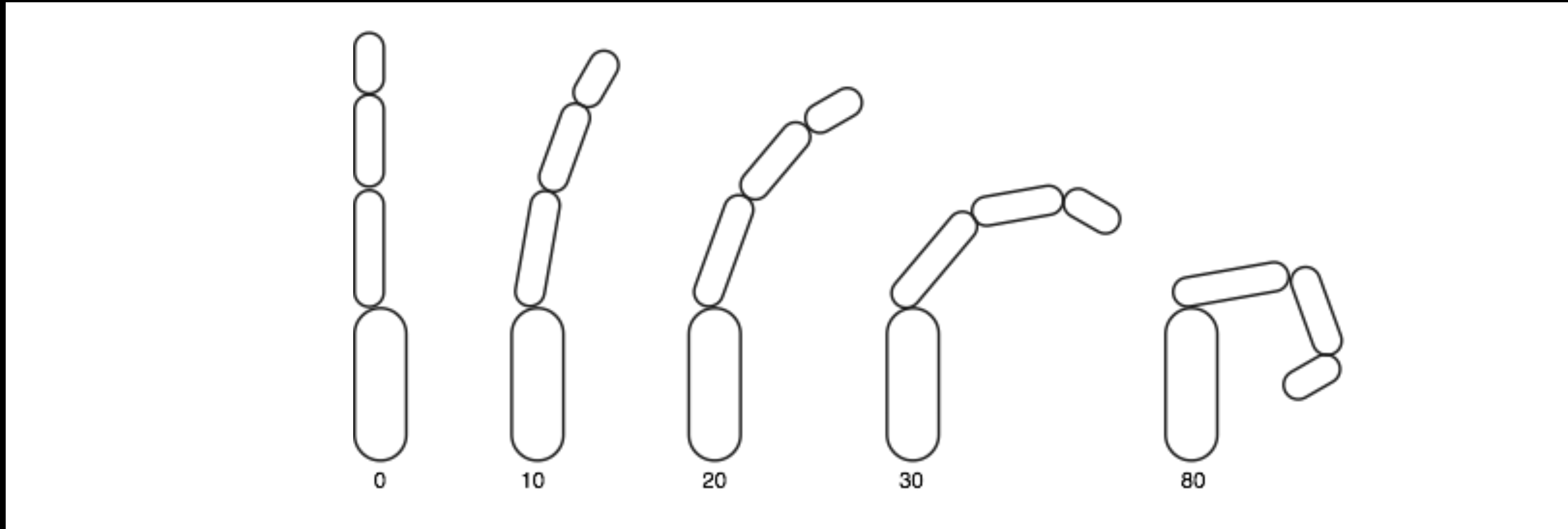


5. Modeling an Hand

- Simple approach
 - Separate movement for each finger

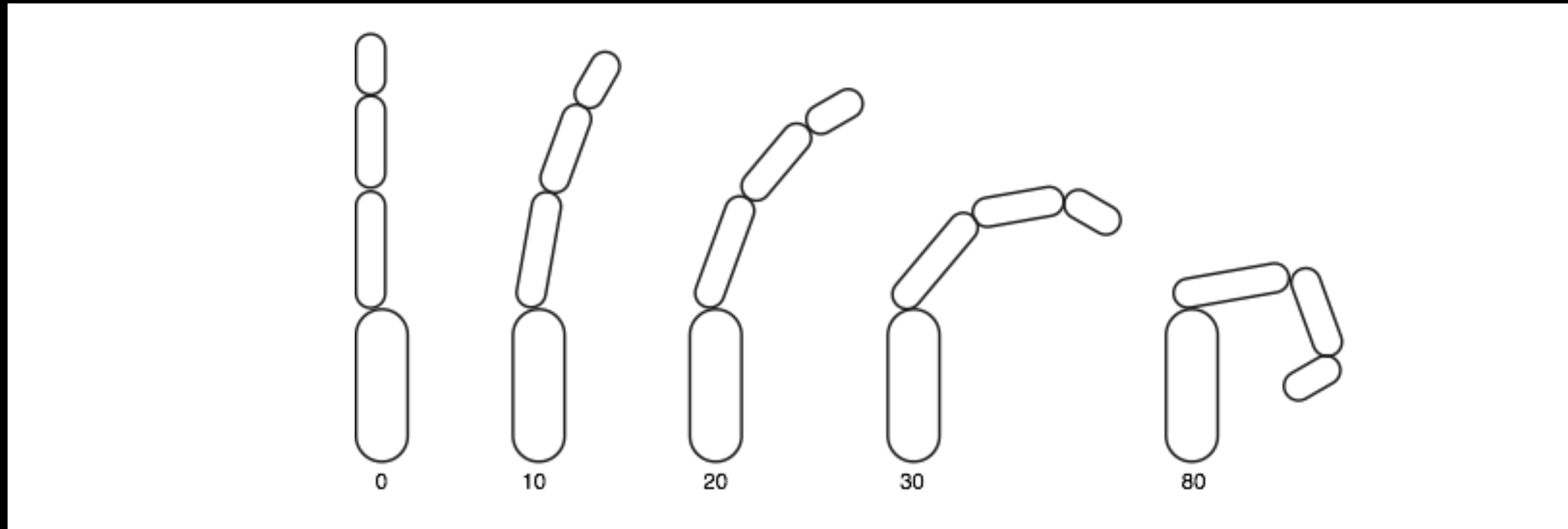
5. Modeling an Hand

- One finger motion to grasp



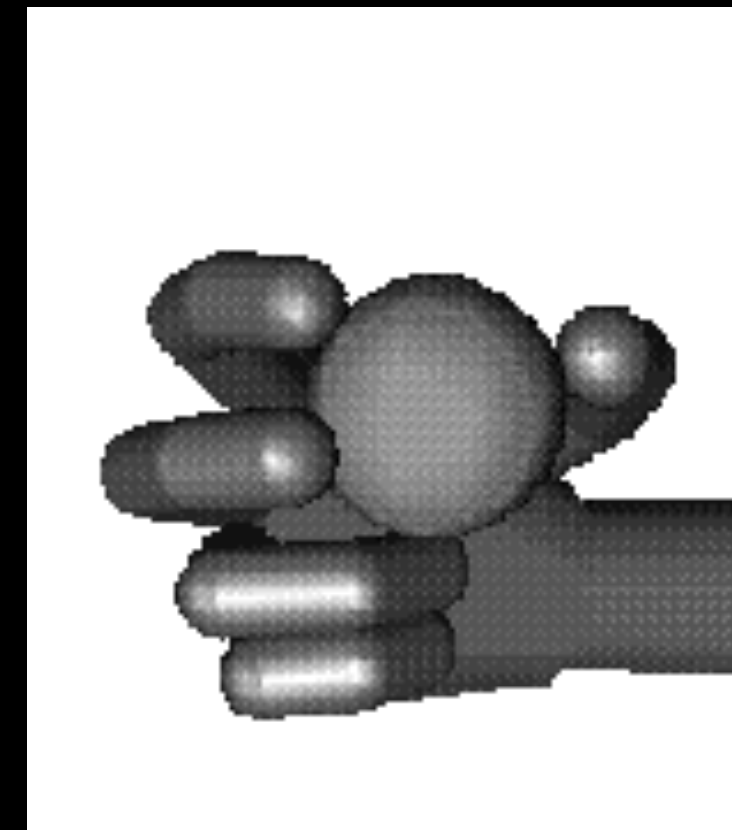
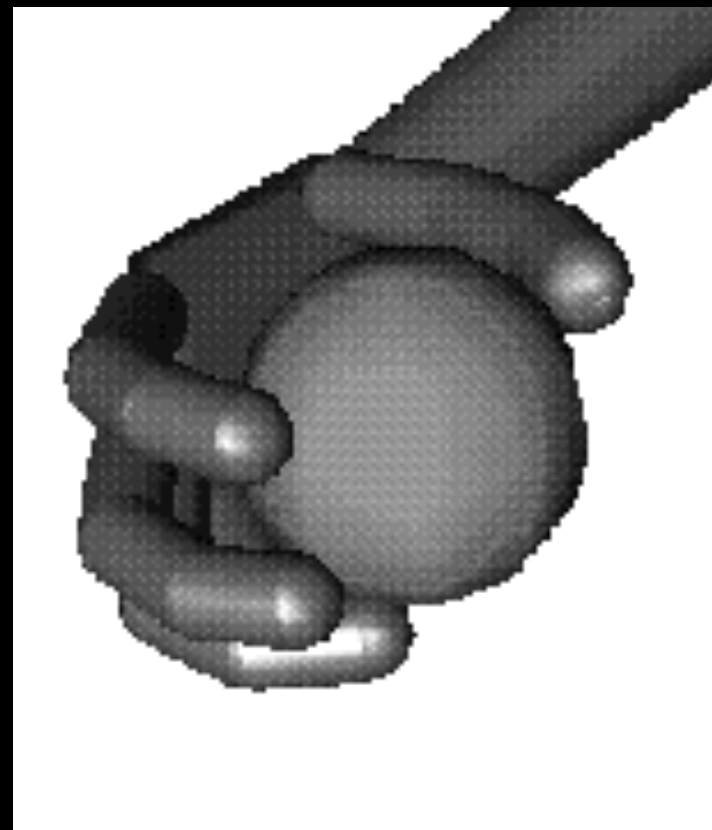
5. Modeling an Hand

- Grasping - gradually attaching finger links to the target

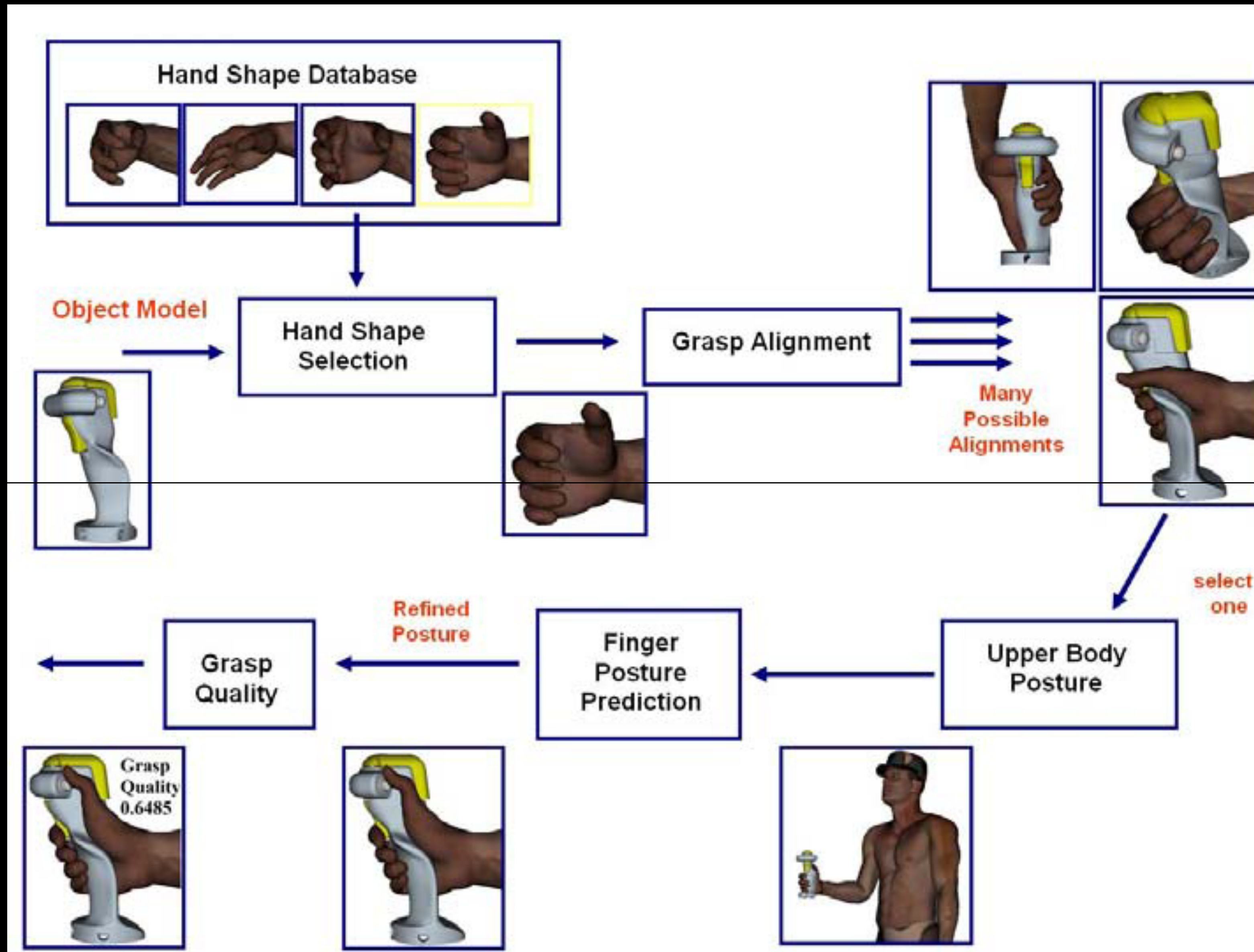


5. Modeling an Hand

- Finger by Finger control to grasp
- Types of grasping

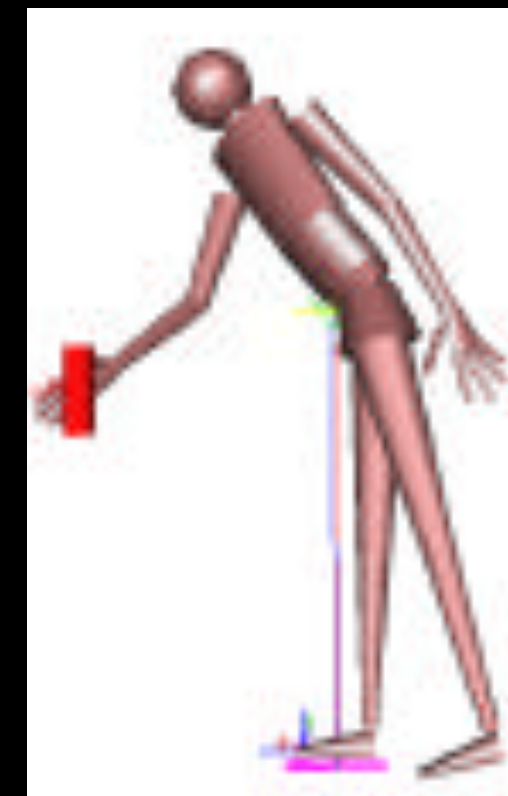


5. Modeling an Hand



Final Touch

- Reaching close object or distant



Summary

- Reaching
 - Most important and simple upper body animation
 - Model of arm, shoulder, and hand
 - Animation control method could be different for arm, shoulder, and hand
 - Further physical model could be considered

Summary

- Human Body Representation
- Rigging and Skinning
- Controlling the human motion example